Blue Data, Boundless Opportunities

How India's seas could power its AI future

BLUE ECONOMY • HUMAN CAPITAL • GLOBAL COMPETITIVENESS

01 An Ocean Within Reach

Stand on any Indian beach at dusk and you'll see it: fishing boats rocking at anchor, tea stalls steaming, families out for an evening walk. What you won't see is the computing power quietly shaping our jobs, our medicine and our classrooms. And yet the two may be closer than we think.

India has the Arabian Sea to the west, the Bay of Bengal to the east and the Indian Ocean to the south, a position few countries can match. While the world races to lead in artificial intelligence, we can ask a different question: **what if our seas helped us win?**

This essay explores one possible answer: coastal and underwater data centres, who could use them, how they might pay their way, and what we must protect along the way.

02 From Coastline to Computing

Data centres are the quiet engines behind our banking apps, video calls and AI tools. They need steady power, serious cooling and fast connections, and coasts offer all three. Undersea cables already come ashore there, and seawater can help carry heat away.

So the plan is a gentle one. Start with a coastal facility, learn from running it, then try a single small underwater pilot: a sealed module, linked to land by power and fibre-optic cables and watched remotely.

Let’s be honest, though. Underwater equipment is hard to repair and costly to maintain. That is exactly why we begin small. A pilot teaches us what works before we spend money on a promise we cannot keep.

> *“Technology becomes powerful only when ordinary people can reach it.”*

03 Computing for Every Classroom

Machines alone don’t build a nation. People do. Picture a first-year student in a small town who can’t afford a powerful laptop. On a platform backed by this infrastructure, she rents computing power for a few hours, pays only for what she uses, and trains her very first model.

Researchers could run experiments the same way, and startups could test bold ideas without buying hardware they can’t afford.

Affordable student plans and targeted support would matter most. So would AI tools in Indian languages, because a farmer in Madurai or a shopkeeper in Guwahati shouldn’t need fluent English to benefit. Get this right and young Indians stop being only users of technology and become its creators. **That, more than any cable or server, is the real return on the investment.**

04 Serving the World, Earning Our Keep

The same facilities could serve companies, developers, universities and startups abroad, who would pay to run AI applications, process data, train models or store information. That income could keep prices low for students and small businesses at home.

Competitiveness isn't only about owning the biggest machines. Countries compete on **talent, cost and trust**. A nation that trains skilled people, charges fairly and keeps data safe will win customers, and customers bring jobs and new companies with them.

Government and private partners would likely share the cost and the risk. Still, the numbers only work if customers truly come, which means reliable service and competitive prices. Neither can be taken for granted.

1
START SMALL
Begin with a coastal facility and one small underwater pilot.

2
OPEN THE DOORS
Give students, researchers and startups computing at fair prices.

3
EARN GLOBALLY
Serve customers abroad so access stays affordable at home.

4
PROTECT FIRST
Study marine life first. Expand only if the ocean is unharmed.

05 The Blue Economy Gets Smarter

Technology should serve the sea itself. Combined with ocean, satellite and weather data, AI can help ports schedule ships, help vessels choose safer routes and give fishing families clearer warnings about rough water. A fisherman who knows the sea before he leaves the shore protects both his catch and his life.

Researchers could use the same computing power to study coral health, fish stocks and warming oceans. These services would also create skilled jobs in coastal towns, where young people so often leave to find work elsewhere.

06 Protecting What Lives Below

Any project that touches the ocean must respect it. Before choosing a site, experts should study local marine habitats and steer clear of coral reefs and breeding grounds. Equipment must be tightly sealed, and cooling designed so the heat released into the water stays within safe limits.

Temperature, noise and marine life should be monitored throughout, and cables laid with care to disturb the seabed as little as possible. Marine scientists should review the pilot honestly before anything expands.

> ***If the ocean pays the price for our progress, we have misunderstood progress.***

07 An Ocean of Opportunity

India doesn't need to copy anyone else's road to artificial intelligence. We can build on what we already have: a long coastline, a young population and a rising digital economy.

Coastal and perhaps underwater computing could make AI more reachable for ordinary citizens, open doors for businesses and offer services to the world. With careful planning, public-private cooperation and real protection for marine life, this idea could help build a country that is more innovative, more inclusive and more competitive.

The sea has always connected India to the world.
It may now connect India to its future.